Cambridge International AS & A Level

**Cambridge International Examinations**
Cambridge International Advanced Subsidiary and Advanced Level

**CHEMISTRY** **9701/21**

Paper 2 AS Level Structured Questions **May/June 2017**

MARK SCHEME

Maximum Mark: 60

**Published**

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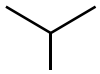
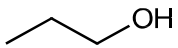
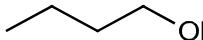
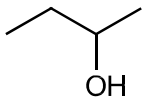
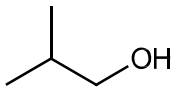
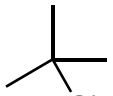
| Question | Answer | Marks |
|---|---|---|
| 1(a) | The mass of a molecule<br>OR the (weighted) average / (weighted) mean mass of the molecules | **1** |
| | Relative / compared to $\frac{1}{12}$ (the mass) of <u>an atom</u> of carbon–12<br>OR on a scale in which a carbon–12 atom / isotope has a mass of (exactly) 12 (units) | **1** |
| 1(b)(i) | **3** | **1** |
| 1(b)(ii) | **8** | **1** |
| 1(b)(iii) | $C_3H_8O + 4½O_2 \rightarrow 3CO_2 + 4H_2O$ | **1** |
| 1(b)(iv) |  <br>**AND** propan–2–ol / 2–propanol | **1** |
| |  <br>**AND** propan–1–ol / 1–propanol | **1** |
| | Alternative answers (any two):<br>  <br>**AND** butan–1–ol / 1–butanol<br>  <br>**AND** butan–2–ol / 2–butanol<br>  <br>**AND** (2–)methylpropan–1–ol / (2–)methyl–1–propanol<br>  <br>**AND** (2–)methylpropan–2–ol / (2–)methyl–2–propanol | |

| Question | Answer | Marks |
|---|---|---|
| 1(b)(v) | correct conversions of data to SI/consistent units<br>p = 100 000 ; V = $20 \times 10^{-6}$ ; T = 393 | **1** |
| | calculation of *n ( = pVIRT)* from M1 values<br>$n = \frac{100 \times 10^3 \times 20 \times 10^{-6}}{8.31 \times 393}$ | **1** |
| | calculation of mass *m ( = n × Mr)* **AND** answer correct to **3sf**<br>*m* = $6.12 \times 10^{-4} \times 60 = 0.0367$ (g)<br><br>Alternative answer for using $C_4H_{10}O$: *m* = $6.12 \times 10^{-4} \times 74 = 0.0453$ (g) | **1** |
| | **Total:** | **10** |

| Question | Answer | | | Marks |
|---|---|---|---|---|
| 2(a) | substance | type of bonding | type of lattice structure | |
| | copper | metallic | giant/metallic | **1** |
| | ice | covalent OR hydrogen(-bonding) / H(–bonding) | hydrogen-bonded / simple / molecular | **1** |
| | silicon(IV) oxide | covalent | giant (molecular) / macromolecular | **1** |
| | iodine | covalent | simple / molecular | **1** |
| | sodium chloride | ionic | giant / ionic | **1** |
| 2(b)(i) | hydrogen bonding | | | **1** |

| Question | Answer | Marks |
|---|---|---|
| 2(b)(ii) | H-bond between O and H of different molecules | **1** |
| | minimum **three** partial charges (in a row) over **two** $H_2O$ molecules,<br>i.e.:<br>either $^{\delta-}O—H^{\delta+}$ - - - - $^{\delta-}O$<br>or $H^{\delta+}$ - - - - $^{\delta-}O—H^{\delta+}$ | **1** |
| | lone pair of electrons on O of H-bond, in line with H-bond | **1** |
| 2(c)(i) | X = liquid AND Z = solid | **1** |
| | Y = liquid and solid OR 'liquid / solid' OR 'liquid OR solid' | **1** |
| 2(c)(ii) | (kinetic) energy reducing | **1** |
| | motion slowing *owtte* | **1** |
| 2(c)(iii) | energy given out / released forming bonds / forming bonds exothermic | **1** |
| | compensates for / counteracts heat loss / cooling *owtte* | **1** |
| | **Total:** | **15** |

| Question | Answer | Marks |
|---|---|---|
| 3(a)(i) | **A** | **1** |
| 3(a)(ii) | **H** | **1** |
| 3(a)(iii) | **G** | **1** |
| 3(a)(iv) | **B** | **1** |
| 3(a)(v) | **F** | **1** |
| 3(b)(i) | (strong) heating | **1** |
| | (to provide / overcome) high activation energy | **1** |
| 3(b)(ii) | white flame / white light / white smoke / white solid | **1** |
| 3(b)(iii) | $Mg(s) + 2H_2O(l) \rightarrow Mg(OH)_2(s) + H_2(g)$ | **2** |
| 3(c)(i) | $2Mg(NO_3)_2 \rightarrow 2MgO + 4NO_2 + O_2$ | **1** |
| 3(c)(ii) | $CaCO_3 \rightarrow CaO + CO_2$ | **1** |
| | $CaO + H_2O \rightarrow Ca(OH)_2$ | **1** |
| 3(d)(i) | reduce acidity in soil / increase pH of soil | **1** |
| | (both) basic / base(s) | **1** |
| 3(d)(ii) | $CaCO_3 + 2H^+ \rightarrow Ca^{2+} + CO_2 + H_2O$<br>OR<br>$CaCO_3 + 2H^+ \rightarrow Ca^{2+} + H_2CO_3$ | **1** |
| | **Total:** | **16** |

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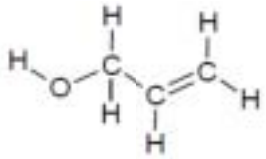
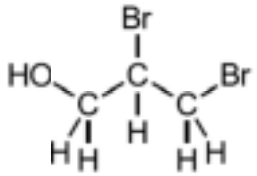
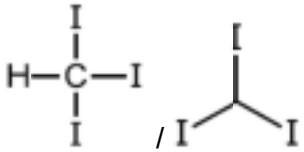
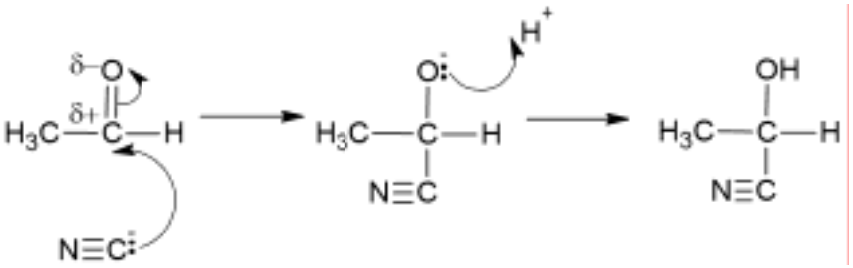
| Question | Answer | Marks |
|---|---|---|
| 4(a)(i) | (molecules / isomers with) the same molecular formula / same number of atoms of each element | **1** |
| | different structural / displayed formulae / arrangement of bonds | **1** |
| 4(a)(ii) | $sp^2$ overlap of (2)s with two (2)p (atomic) orbitals | **1** |
| | $sp^3$ overlap of (2)s with all three (2)p (atomic) orbitals | **1** |
| 4(a)(iii) | $sp^2 = 116° – 124°$ | **1** |
| | $sp^3 = 106° – 112°$ | **1** |
| 4(b)(i) |  | **1** |
| 4(b)(ii) | (electrophilic) addition | **1** |
| | bromine decolourises / turns colourless / fades (from orange / brown) | **1** |
| 4(b)(iii) | $HOCH_2CHBrCH_2Br$ OR  | **1** |
| 4(b)(iv) | $CO_2$ / carbon dioxide | **1** |
| 4(c)(i) | **P** = propanal | **1** |
| | **Q** = propanone | **1** |

| Question | Answer | Marks |
|---|---|---|
| 4(c)(ii) | tr(i)iodomethane / $CHI_3$ /  /  | **1** |
| 4(d)(i) | (molecules / isomers with) the same (molecular and) structural formula | **1** |
| | Any two of:<br>chiral centre / C attached to four different groups / atoms<br>non-super(im)posable mirror images<br>different spatial / 3D arrangement of atoms (owtte)<br>different rotation of plane-polarised light | **1** |
| 4(d)(ii) |  | |
| | curly arrow from lone pair on :C≡N to $C^{(\delta+)}$ | **1** |
| | correct dipole on carbonyl $^{\delta+}C=O^{\delta-}$ AND curly arrow from bond to $O^{(\delta-)}$ | **1** |
| | correct intermediate, including $C–O^-$ AND curly arrow from lone pair to $H^+$ | **1** |
| | **Total:** | **19** |